

BHUVESH KUMAR

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SUMMARY

Software Engineer with 3.5 years at Oracle Singapore owning quality engineering and AI tooling across Java Management Service (v6.0–v11). Self-initiated and shipped 3 enterprise AI tools on OCI — a RAG pipeline (60% faster test spec generation), a computer vision redaction system (85% manual effort reduction), and an open source MCP server (PR #175, [oracle/mcp](https://github.com/oracle/mcp)). Now pursuing MTech in AI Systems at NUS-ISS full-time, targeting AI/ML engineering roles where production rigour matters.

SKILLS

Languages: Java, Python, SQL, TypeScript, C#, .NET, Shell Scripting

LLMs & Agents: RAG Pipelines, LLM Orchestration, Agentic AI, MCP, LLM Evaluation (LLM-as-judge, HITL), Prompt Engineering, LangChain, FastMCP

ML Core: Supervised & Unsupervised Learning, Ensemble Methods, Feature Engineering, Recommender Systems, XGBoost, scikit-learn

Deep Learning: PyTorch, TensorFlow, CNNs, Transfer Learning, Object Detection, Image Classification

NLP: Transformers, Hugging Face, Text Classification, NER, Small Language Models

Frameworks: FastAPI, Flask, Spring Boot, Helidon SE, Neo4j, LangChain, Dash, OR-Tools

AI Services: OCI GenAI, OCI Vision, OCI AI Agents, Gemini, OCI GenAI Agent

Infrastructure: Oracle Cloud (OCI), AWS, Docker, Jenkins, CI/CD, Git, Grafana, Canary Deployments

Currently: LangGraph, LoRA / QLoRA, RAGAS, Responsible AI, Multimodal Models

EXPERIENCE

Software Engineer | Oracle | Singapore *Oct 2022 – Apr 2026*

Quality-focused Software Engineer on Java Management Service — owning test coverage across 6 major releases (v6.0–v11) and self-initiating 3 AI tools that went into production on OCI.

- Owned end-to-end test coverage for JMS platform features: Dynamic and Static Java Library Scanning (SCA), Vulnerability Scan and Risk Scoring, Automatic Java Update distribution, Performance Analysis Reports (JDK Flight Recording), Java Migration Analysis Reports, and Deployment Rule Set (DRS) management.

- Designed and maintained automated test infrastructure with canary deployments and Grafana-based alerting for regression detection before release rollout.
- Self-initiated and shipped JMS AI Toolkit — a 5-stage RAG pipeline (OCI GenAI Agent + Gemini) with citation traceability, LLM-as-judge evaluation, and HITL review; reduced test spec creation time by 60% for the SQE team.
- Built Sensitive Data Redaction pipeline (Helidon SE + OCI Vision + OCI GenAI) for automated PII masking in LiveLabs content; reduced manual effort by 85%.
- Contributing to oracle/mcp (public Oracle GitHub, 316 stars): open source OCI JMS MCP Server (PR #175) — 13 tools covering fleet health, compliance, security notices, and inventory; 55 unit tests at 90% coverage, OCA signed.
- Proposed and delivered PoC for JMS LiveLabs Generator — OCI GenAI RAG system ingesting PRDs, Figma files, and meeting notes to generate tutorials via chat; PoC demoed to leadership.
- Participated in feature design discussions, sprint planning, and cross-functional collaboration across 4 global regions (Singapore, India, Europe, US) with SRE, SQE, dev, and documentation teams.

AI Engineer (Volunteer) | better.sg | Singapore Mar 2026 – Present

- Building AI-powered tools for Singapore nonprofits applying LLM pipelines, FastAPI backends, and automation to real social-sector problems.
- Applying production engineering practices — design, build, deploy — to deliver working software with cross-functional volunteer teams.

Software Engineer Intern | Oracle | Singapore Mar 2022 – Jul 2022

- Built and published Oracle LiveLabs tutorials for OCI, JMS, and network configurations used by internal and external engineers globally.
- Automated internal software release processes using scripting and CI/CD pipelines, improving release efficiency.

Validation Engineer | Exicom Tele-Systems Limited | Gurgaon, India Jul 2016 – Nov 2019

- Developed automated test setups and Python scripts for validating electronic devices and EV chargers; improved production quality through data-driven root cause analysis.

PROJECTS

JMS AI Toolkit: RAG-Based Test Specification Generator | Oracle · Enterprise-Deployed · 2025

- 5-stage LLM pipeline (LLM0–LLM5+N) using OCI GenAI Agent for retrieval and Gemini for generation; citation traceability (Citation → Fact → Requirement → TA → Test) enforced at every stage.

- LLM-as-judge evaluation scoring output quality per stage; HITL review gates (Auto / HITL / SuperHuman workflows); citation auto-validation and cross-batch duplicate detection.
- 60% reduction in test specification creation time for the Oracle JMS SQE team. SSE streaming for real-time progress. Session persistence to OCI Object Storage. DOCX export.

OCI JMS MCP Server | Oracle · Open Source · 2026 | PR #175 · oracle/mcp

- Python MCP server contributing to Oracle's public oracle/mcp repository (316 stars); 13 read-only JMS tools covering fleet health summaries, detailed diagnostics, service notices, Java runtime compliance, inventory, and plugins.
- 55 unit tests passing at 90.04% coverage (meeting Oracle's required 90% threshold); 10 BDD e2e scenarios via Behave with 40 steps; OCA signed and verified.

Sensitive Data Redaction in Images | Oracle · Enterprise-Deployed · 2024

- Helidon SE pipeline using OCI Vision (object detection) and OCI GenAI (context-aware masking) to automatically detect and redact PII from images before LiveLabs publication.
- 85% reduction in manual effort; deployed internally across Oracle teams.

HDB ResaleXAI: Explainable AI Dashboard | NUS-ISS · IRS Module Project · 2026

- End-to-end explainable AI system for Singapore HDB resale price prediction: XGBoost with SHAP and LIME explanations, React dashboard, FastAPI backend, Neo4j knowledge graph, and Gemini-powered RAG conversational interface.
- PyTorch learned block embeddings; XGBoost benchmarked against Random Forest and LightGBM on RMSE, MAE, and inference latency; normalised ensemble model to address post-2022 price drift.

Agentic UI Test Navigator | Oracle · Research & PoC · 2025

- ReAct-pattern agentic system on OCI GenAI autonomously navigating the JMS web UI using 6 tools (Login, Map Elements, Navigate, Click, Screenshot, Wait); 39-step reasoning trace (THOUGHT → DECISION → OBSERVATION), mapping 238 interactive elements per page.

Explainable AI Web Application | A*STAR Research Collaboration · Final Year Project · 2020–2021

- Flask + Dash web application where users upload any scikit-learn model and dataset to generate interactive XAI dashboards (SHAP, LIME, ICE, surrogate decision trees, what-if analysis) without writing code. Built with A*STAR researchers. 1,844-line production codebase.

EDUCATION

MTech in Artificial Intelligence Systems *Jan 2026 – Dec 2028*

National University of Singapore (NUS-ISS) | Full-time from Apr 2026 **In Progress**

Coursework: Computer Vision, Deep Learning, NLP, Explainable & Responsible AI, Knowledge Graphs, Agentic AI Solutions, MLSecOps/LLMSecOps, AI & Cybersecurity, Recommender Systems.

Graduate Diploma in Systems Analysis *Jul 2021 – Aug 2022*

National University of Singapore (NUS-ISS) **GPA 4.92/5 | Distinction | ISS Prize (2nd best)**

PG Certificate in Data Science & ML *2021 – 2022*

IIT Roorkee | Part-time

BSc Computer Science *2019 – 2021*

Coventry University (PSB Academy, Singapore) **First Class Honours | Top graduate**

CERTIFICATIONS

- ★ **OCI Generative AI Professional** — Oracle · OCI Academy
- ★ **AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents** — Udemy · Ed Donner
- ★ **Vector Databases for RAG: An Introduction** — IBM
- Neural Networks and Deep Learning — Coursera · deeplearning.ai
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization — Coursera · deeplearning.ai
- LLM Engineering: Master AI, Large Language Models & Agents — Udemy
- Python for Data Science and Machine Learning Bootcamp — Udemy

COMMUNITY & LEADERSHIP

- AI Engineer (Volunteer) — better.sg · Mar 2026–present · building LLM tools for Singapore nonprofits
- Volunteer — Willing Hearts Soup Kitchen · 2025–present · weekly meal programme
- Vice President, AI Focus Group — Singapore Computer Society · 2020–2021
- Product Designer (External) — Google Developer Student Clubs NUS · 2021–2022
- Member — Singapore Computer Society · 2025–present